CS 4410: Midterm Summer 2026B

Exam Time: 50 Minutes ***Total Marks: 30***

Write Your section, your Troy ID, and your name on every sheet.

Troy ID: **Name:**

Q1 [10 points] Answer any two of the following. Each response is worth 5 points.

(A) Distinguish the likelihood $p(\mathcal{X} \mid \theta)$ from the posterior $p(\theta \mid \mathcal{X})$. Why is the log-likelihood commonly optimized instead of the original likelihood product?

(B) Distinguish a validation set from a test set. Explain how high bias and high variance appear in training and validation performance.

Part II: Numerical and Problem-Solving Questions **20 points**

Q2 [4 points] A disease has prevalence $P(D) = 0.01$. A diagnostic test has sensitivity

$$P(+ \mid D) = 0.80$$

and false-positive rate

$$P(+ \mid \overline{D}) = 0.096.$$

A randomly selected person receives a positive result. Compute $P(D \mid +)$ using Bayes' rule. Show all steps and briefly interpret the result.

Q3 [4 points] Assume that the observations

$$\mathcal{X} = \{2, 3, 5, 6, 9\}$$

are independently drawn from a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$. Compute the maximum-likelihood estimates

$$\widehat{\mu}_{ML} \qquad \text{and} \qquad \widehat{\sigma}^2_{ML}$$

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Part I: Short-Answer Questions **10 points**

Q1 [10 points] Answer following. Each response is worth 5 points.

(A) Explain the meanings of the specific boundary S, the general boundary G, and the version space in Candidate Elimination. What does an empty version space indicate?

(B) List the major stages of a complete machine-learning project, from defining the problem to presenting the solution.

Part II: Numerical and Problem-Solving Questions **20 points**

Q2 [7 points] Apply the Candidate Elimination algorithm to the training data below. Show the S and G boundaries after every example and state the final version space.

Weather	**Temperature**	**Wind**	**Class**
Sunny	Warm	Weak	+
Rainy	Warm	Weak	+
Cloudy	Cold	Strong	−
Sunny	Cold	Weak	+

Q3 [7 points] Consider two classes C_1 and C_2 with the following loss matrix:

	C_1	C_2
α_1 : choose C_1	0	5
α_2 : choose C_2	2	0

Let

$$P(C_1 \mid x) = p, \qquad P(C_2 \mid x) = 1 - p.$$

(a) Write $R(\alpha_1 \mid x)$ and $R(\alpha_2 \mid x)$.

(b) Derive the decision threshold without rejection.

(c) Add a reject action α_R with loss 0.6 for either true class. Determine the ranges of p for choosing C_1, choosing C_2, and rejecting.

Q4 [6 points] Consider the feature values

$$X = \{10, 20, 30, 40, 100\}.$$

For $x = 40$, compute:

(a) Min–Max scaling,

$$x' = \frac{x - X_{\min}}{X_{\max} - X_{\min}};$$

(b) robust scaling when the median is 30, $Q_1 = 20$, and $Q_3 = 40$,

$$x' = \frac{x - \text{median}(X)}{Q_3 - Q_1};$$

(c) Max-Abs scaling.

Which of these methods is generally least influenced by the outlier 100, and why?

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